

Redundant Power Selector Secondary Power Malfunction

Symptoms

- Primary power supply failure error message shown on diesel control unit (DCU).

How To Use This Tree

This symptom tree can be used to troubleshoot signal from the speed pick-up is signal lost. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Secondary power failure error message.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box (C.I.B.) wiring.	
	STEP 1A. Check the control panel display for faults.	Control panel indicates fault(s)?
STEP 2.	Check redundant power selector wiring.	
	STEP 2A. Check redundant power selector secondary power input SUPPLY wire for voltage +18 VDC.	Less than +18±0.2 VDC?
	STEP 2B. Check redundant power selector primary power input SUPPLY wire for voltage +18 VDC.	Less than +18±0.2 VDC?
STEP 3.	Check redundant power selector voltage.	
	STEP 3A. Check redundant power selector power output SUPPLY.	Output within ± 0.5 VDC?

STEP 1. Check the customer interface box (C.I.B.) wiring.

STEP 1A. Check the control panel display for faults.

Conditions :

- Locate the control panel display.

Action	Specification/Repair	Next Step
Check the control panel display for faults.	Control panel indicates fault(s)? YES Repair : Troubleshoot the appropriate fault code. Reference the Marine Auxiliary QSB7-DM CM850 Fault Code Troubleshooting Manual, Bulletin 4325972, Section TF, or the ISM and QSM 11 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666266, Section TF, X15 CM2350 X125M Fault Code Troubleshooting Manual, Bulletin 5504346.	Go to appropriate fault code troubleshooting tree.
	Control panel indicates fault(s)? NO Repair : Replace the drive coupling.	2A

STEP 2. Check redundant power selector wiring.

STEP 2A. Check redundant power selector secondary power input SUPPLY wire for voltage $+18 \pm 0.2$ VDC.

Conditions :

- Open the C.I.B.
- Test the redundant power selector secondary power input SUPPLY wire pin 6.

Action	Specification/Repair	Next Step
Check the voltage at pin 6 of the redundant power selector. • Place one test lead at the secondary power SUPPLY wire at pin 6 of the redundant power selector. • Place the other test lead on the panel ground wire at pin 7. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +18±0.2 VDC? YES Repair : Check the batteries. See equipment manufacturer service information.	Repair complete.
	Less than +18±0.2 VDC? NO	2B

STEP 2B. Check redundant power selector primary power input SUPPLY wire for voltage +18±0.2 VDC.**Conditions :**

- Open the C.I.B.
- Test the redundant power selector secondary power input SUPPLY wire pin 1.

Action	Specification/Repair	Next Step
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Check the voltage at pin 1 of the redundant power selector. - Place one test lead at the primary power SUPPLY wire at pin 1 of the redundant power selector. - Place the other test lead on the panel ground wire at pin 2. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +18±0.2 VDC? YES	3A
	Less than +18±0.2 VDC? NO	3A

STEP 3. Check redundant power selector voltage.

STEP 3A. Check redundant power selector power output.

Conditions : - Open the C.I.B. - Test the redundant power selector power output SUPPLY wire pin 17.		
Action	Specification/Repair	Next Step

Check the output voltage at pin 17 of the redundant power selector. - Place one test lead at the power output wire at pin 17 of the redundant power selector. - Place the other test lead on the panel ground wire at pin 16. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Output voltage ± 0.5 VDC of the primary and secondary input voltages? YES	Repair complete.
	Output voltage ± 0.5 VDC of the primary and secondary input voltages? NO Repair : Replace redundant power selector.	1A